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Examining AI-Blockchain Adoption in Life Insurance through an Integrated HOT-TOE Framework --Manuscript Draft--

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Abstract:	This study investigates the drivers and barriers influencing the Actual Adoption of Artificial Intelligence (AsI) and Blockchain technologies in the life insurance sector of Bangladesh. Integrating the Human–Organization–Technology (HOT-fit) and Technology–Organization–Environment (TOE) frameworks, a comprehensive conceptual model was developed to explain adoption behavior. Data were collected through a deductive questionnaire survey from 394 employees across 32 life insurance companies covering all eight divisions of Bangladesh using cluster sampling. The proposed hypotheses were examined using Partial Least Squares Structural Equation Modeling (PLS-SEM). The findings reveal that Organizational Readiness is the strongest predictor of Actual Adoption, primarily driven by Technology Readiness and Business Model Readiness. The positive association between readiness and adoption is strengthened by Perceived Cost considerations and attenuated by Stickiness to Legacy Systems. Notably, system integrity did not significantly predict readiness, suggesting that in compliance-driven markets security functions as an assumed regulatory baseline rather than a distinguishing driver of adoption. This study contributes to the literature by integrating HOT-fit and TOE in the context of AI-Blockchain adoption within life insurance in a developing economy and introduces novel constructs such as Stickiness to Legacy Systems and sustainability-oriented interoperability and cost-benefit perspectives.
Additional Information:	
Question	Response
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access to and citations of your research prior to publication. Telematics and Informatics R offers a free service to post your paper on SSRN, an open access research repository, at the time of submission. Once on SSRN, your paper will benefit from early registration with a DOI and early dissemination that facilitates collaboration and early citations. It will be available free to read regardless of the publication decision made by the journal. This will have no effect on the editorial process or outcome with the journal. Please consult the SSRN Terms of Use and FAQs.	
To complete your submission you must select a statement which best reflects the availability of your research data/code. IMPORTANT: this statement will be published alongside your article. If you have selected "Other", the explanation text will be published verbatim in your article (online and in the PDF). (If you have not shared data/code and wish to do so, you can still return to Attach Files. Sharing or referencing research data and code helps other researchers to evaluate your findings, and increases trust in your article. Find a list of supported data repositories in Author Resources, including the free-to-use multidisciplinary open Mendeley Data Repository.)	Data will be made available on request.

Response to Reviewers

Reviewer 1

Unlike Figure 1, Figure 2 appears to be blurred making text in the figure difficult to read. Please re-attach a high-resolution image for figure 2. The authors acknowledge that figure resolution improvements are "flagged for production." Figure 3 (PLS-SEM model) and Figure 1 (Conceptual Framework) are central to a quantitative methodology paper. The authors should provide publication-quality vector or high-DPI raster exports of these figures before final submission, not defer this to the production team.

The blurred figure has been removed. It showed only the research-design flowchart, which carried generic procedural content and was the least readable of the three, so deleting it was cleaner than re-rendering it. The two figures that matter, the Conceptual Framework and the PLS-SEM model, are now supplied at 300 DPI or higher. With the flowchart gone, the PLS-SEM model becomes Figure 2, and every figure number and in-text reference has been updated to match.

With regard to HTMT discriminant validity, the theoretical justification given is not sufficient. The HTMT values obtained OR-AA (0.945) and TR-OR (0.920) are both exceeding the conservative threshold of 0.90. [...] This logic is circular: if two constructs are so theoretically close that high HTMT is anticipated by design, the question arises whether they are truly distinguishable in this sample. The manuscript does not report confidence intervals for the HTMT ratios. [...] the authors are encouraged to (a) report bootstrapped 95% confidence intervals for these two HTMT ratios and (b) conduct a confirmatory factor analysis comparing a two-factor model against a one-factor model for the OR-AA pair, or alternatively provide item-level discriminant validity evidence via the AVE-correlation comparison (Fornell-Larcker criterion) as a supplementary check.

The circular argument is gone, replaced by both tests the reviewer named.

For (a), we computed bias-corrected bootstrap 95% confidence intervals with 5,000 resamples. The relevant criterion for HTMT is whether the interval excludes 1.00 (Henseler et al., 2015). Both do: OR-AA is [0.886, 0.994] and TR-OR is [0.854, 0.981], and neither upper bound reaches unity.

For (b), we ran the confirmatory factor analysis. A two-factor model treating OR and AA as separate correlated constructs fits significantly better than a one-factor model collapsing all ten indicators into one ($\Delta\chi^2 = 11.3$, $\Delta df = 1$, $p < 0.001$). The two-factor fit is good on its own terms ($\chi^2 = 72.1$, $df = 34$, $RMSEA = 0.053$, $CFI = 0.978$) and better than the one-factor fit ($\chi^2 = 83.4$, $df = 35$, $RMSEA = 0.059$). The two constructs are therefore separable. Their inter-factor correlation of 0.943 shows they remain strongly related, which is what one expects of readiness and the adoption it precedes.

These results appear in Section 4.1.2 with two new tables. On one point of framing, the HTMT inference test is evaluated against 1.00 rather than 0.90; measured that way, both pairs demonstrate discriminant validity. The confidence intervals and the CFA were run in R (semnir and lavaan) from the raw data so the analysis is fully reproducible.

Common Method Bias Remediation: The revised manuscript adds Harman's single-factor test (31.4%) and a full collinearity VIF check (all < 3.3) to address common method variance (CMV). [...] Both diagnostics are

widely recognized as limited. [...] The authors can do the following: (a) acknowledge this as a substantive limitation and moderate claims accordingly, or (b) apply a marker-variable approach by including a theoretically unrelated construct in the model and using it as a covariate, if the dataset permits. At minimum, the limitations section should explicitly note that CMV cannot be ruled out with post-hoc tests alone.

We did both (a) and (b). Alongside the existing Harman's and VIF checks, we added a marker-based assessment using the Lindell and Whitney (2001) approach. Management Support served as an ad-hoc marker, since it had the lowest average correlation with the other constructs (median $|r| = 0.042$). The implied method variance came to roughly 4%, and adjusting for it moved the substantive correlations by no more than 0.013, for instance OR–AA shifted from 0.785 to 0.776. Every relationship kept its magnitude and significance. It is also worth noting that two constructs drawn from the same instrument, Management Support and Stickiness to Legacy Systems, correlate near zero with the rest of the model. Shared method variance would not leave that pattern intact. Separately, we softened the earlier sentence that read the VIF result as confirming the absence of common method bias, and we added a limitation stating plainly that post-hoc tests cannot rule CMV out and that our marker was ad-hoc rather than designed for the purpose.

Regarding Employee-Level Measurement of Organizational Constructs. The manuscript measures constructs such as Organizational Readiness, Business Model Readiness, and Management Support through individual employee responses. [...] Critically, the manuscript does not report the aggregation procedure. [...] The authors must clarify the level of analysis at which PLS-SEM was applied. [...] If individual-level, this should be explicitly acknowledged as a limitation, and any claims about organizational-level adoption should be appropriately hedged.

The analysis is at the individual level. $N = 394$ individual respondents, with no aggregation to the firm. We have said this directly in the manuscript and removed the earlier "aggregated across roles" wording, which described a step we did not perform. The constructs are now framed as individual employees' perceptions of organizational conditions rather than as objective firm-level states. A limitation notes that the findings should be read at the level of individual perception, and that a multi-level design with agreement statistics such as rwg or ICC would be needed to justify firm-level aggregation. The firm-as-actor phrasing in the Discussion has been hedged to match.

About Perceived Cost — Construct Specification and Moderation Interpretation. The Perceived Cost (PC) construct contains only two indicators. PC-1 loading is at 1.000. [...] a unit loading generally means that PC-1 and the latent PC construct are identical. [...] Also, Table 5 reports both a direct effect of PC on AA ($\beta = 0.180$, $p < 0.001$) and a significant moderation term ($PC \times OR \rightarrow AA$, $\beta = 0.053$, $p = 0.011$). [...] a simple slopes analysis or a Johnson-Neyman floodlight analysis would substantially strengthen this section.

The 1.000 was a table error, not a real estimate. When we rechecked the output, the loadings column in Table 4 had slipped down by one row, and a value of 1.000 that belongs to the single-indicator $PC \times OR$ interaction term (where a unit loading is expected by construction) had landed on PC-1. The true loadings are $PC-1 = 0.884$ and $PC-2 = 0.887$. Perceived Cost is an ordinary two-indicator reflective construct ($AVE = 0.786$, composite reliability = 0.880), so the redundancy concern does not arise once the table is corrected.

On the interpretation, we added both a simple slopes analysis and a Johnson-Neyman analysis in Section 4.2.1. The effect of Perceived Cost on Actual Adoption is positive and significant at every level of readiness, and it grows as readiness rises: the slope is 0.124 at one standard deviation below the mean ($p = 0.006$), 0.178 at the mean ($p < 0.001$), and 0.232 at one standard deviation above ($p < 0.001$). The Johnson-Neyman result shows the effect stays significantly positive across effectively the whole observed range of readiness. So the moderation strengthens the cost effect rather than reversing it.

Regarding Mediation Analysis Confidence Intervals Contain Zero. Table 7 reports a significant negative indirect effect for $SLS \rightarrow OR \rightarrow AA$ ($\beta = -0.077$, $CI [-0.134, 0.013]$, $p = 0.034$). [...] the reported 95% confidence interval includes zero ($0.013 > 0$) [...] The reported p-value and the confidence interval are therefore contradictory. [...] The authors should re-run the bootstrapping procedure (5,000 resamples, bias-corrected accelerated intervals) and report corrected values. The same issue should be checked for all indirect paths in Table 7.

The contradiction came from the table being filled with percentile intervals while our stated method, and the correct one for indirect effects, is bias-corrected. We rebuilt Table 7 and the accompanying text from the bias-corrected output, again with 5,000 resamples. Under these intervals the $SLS \rightarrow OR \rightarrow AA$ effect is $[-0.143, -0.007]$, which excludes zero and agrees with its p-value. We then checked every indirect path against the output: seven of the eight are significant with intervals clear of zero, and $SI \rightarrow OR \rightarrow AA$ stays non-significant at $[-0.007, 0.130]$, which lines up with the non-significant $SI \rightarrow OR$ direct path.

f^2 Values in Table 5. The f^2 value for $POB \rightarrow OR$ is reported as 0.825. Thos would represent a very large effect size under Cohen's guidelines ($\geq 0.35 = \text{large}$). This appears potentially erroneous because the standardized path coefficient is only $\beta = 0.161$. Please verify this value against the SmartPLS 4 output and correct if necessary.

We checked it against the f-square output and the reviewer is right that it was wrong. The correct value is 0.033, a small effect under Cohen's benchmarks. Table 5 has been corrected and the value added to the effect-size discussion in Section 4.2.1.

Regarding Research Questions and Framework Novelty. RQ1 asks which theoretical framework "offers the most comprehensive perspective." As written, this is a question the authors answer by assertion (they chose HOT-fit + TOE) rather than through systematic comparison. Either reframe RQ1 as a statement of rationale, or acknowledge that the study does not empirically compare competing frameworks.

RQ1 has been reframed. Rather than asking which framework is most comprehensive, which implies a comparison we did not run, it now asks how the integrated framework's factors shape readiness and adoption, which is what the study tests. Section 2.1 states that the integration is a deliberate theoretical choice and says outright that the study does not empirically compare competing frameworks. The answer to RQ1 in the Discussion has been aligned with the new wording.

Non-significant System Integrity Path. The discussion of the null $SI \rightarrow OR$ result offers an interesting contextual explanation (that system integrity is treated as a regulatory baseline in Bangladesh). This is the study's most substantive null finding and deserves more prominent treatment in both the Discussion and the Conclusion, where it currently receives minimal attention.

We agree it was underplayed and have given it room. The Discussion now spells out the mechanism: under IDRA compliance, system integrity reads as a mandated baseline that every system must meet, not as something that sets more-ready organizations apart from less-ready ones, which is why it does not predict readiness even though it is blockchain's defining feature. We tie this to the matching non-significant indirect effect, and we treat the finding directly in the abstract, in Section 5.1 as a boundary condition for the HOT-fit technology dimension, and in the Conclusion.

Generalizability Claims. The abstract and conclusion present findings as informative for "other developing economies." The sample is limited to one industry in one country. [...] the discussion of transferability should explicitly identify which contextual features of Bangladesh (regulatory environment, level of InsurTech maturity, workforce digital literacy) would need to be similar for the findings to apply elsewhere.

The claim has been narrowed from developing economies as a category to conditional transfer. The Conclusion and Limitations now name the three features another setting would need to share for the findings to carry: a regulatory environment where security operates as a compliance baseline rather than an incentive, an early stage of InsurTech maturity, and a still-developing level of workforce digital literacy. We note that settings differing on these may behave differently, and that transfer should not be assumed across developing economies as a group. The three features are the ones the reviewer listed.

Citation Consistency. Despite the stated correction in response to R1.2, several citation inconsistencies persist. [...] references [68] and [69] [...] do not appear to match the descriptions used in the text. [...] the reference [71] Chin WW. The Partial Least Squares Approach to Structural Equation Modeling. 2015. is incomplete. [...] Authors should carefully cross-check all numbered citations against their in-text usage.

The reference list has been rebuilt and checked with our co-author's reference manager, the malformed and incomplete entries corrected, and the numbered citations reconciled against their use in the text.

The authors can strengthen the discussion on uneven adoption of Blockchain in general by referring and citing a recommended reference.

We appreciate the suggestion. Having looked at the recommended reference, and given the Editor-in-Chief's note that recommended references may be set aside, we have kept our current citations, which we think support the discussion adequately.

Reviewer 4

Please improve the literature review by providing a more critical discussion of previous studies and clearly highlighting the research gap and novelty of the proposed work.

The literature review is built to arrive at a specific gap rather than to catalogue prior work. Section 2.1 sets out why neither HOT-fit nor TOE is sufficient alone, naming what each leaves out. Section 2.2 closes on a three-part gap: no integrated HOT-fit–TOE model of AI-blockchain adoption in insurance; almost no empirical evidence from South Asian emerging markets such as Bangladesh; and little treatment of perceived cost as a moderator of the readiness-to-adoption path. The novelty sits in the same place, the integrated framework, the introduction of Stickiness to Legacy Systems as a separate barrier construct, and perceived cost modeled as a moderator. We have made that gap-and-novelty passage more prominent so it is easy to find.

Expand the methodology section by providing clearer justification for the selected sampling technique and statistical analysis.

Both are strengthened. On sampling, we explain that no public individual-level frame of insurance employees exists in Bangladesh, which rules out simple random sampling and makes cluster sampling by firm and role both necessary and a way to cover ownership types, divisions, and functions. On analysis, we added an explicit case for PLS-SEM over covariance-based SEM: the model is large (eleven constructs, forty-seven indicators, a moderation term), the aim is to explain variance in readiness and adoption, and PLS-SEM handles interaction terms directly. This is in Section 3.

The discussion section can be strengthened by comparing the findings with more recent studies and explaining their practical significance.

We deepened the comparisons. The system integrity null is set against the InsurTech literature that treats integrity as a driver, with a reason for the divergence. The perceived-cost moderation is compared with prior work on cost as a moderator of blockchain adoption, and we note our result extends it by showing the interaction amplifies rather than dampens. The dominance of Organizational Readiness is related to earlier readiness-adoption work. Practical significance is now attached to individual findings through the Discussion, on top of the managerial implications already in Section 5.3.

Please provide a clearer explanation of the reliability and validity results, especially for constructs with lower reliability values.

The low-alpha construct, Management Support ($\alpha = 0.613$), is handled in Section 4.1.1. Cronbach's alpha understates reliability for a two-item construct because it assumes tau-equivalence and is sensitive to scale length. The variance-based measure, composite reliability, clears 0.70 at 0.757, and the construct's AVE of 0.707 is the highest in the model. On that basis the construct is measured reliably and validly.

Expand the limitations section by discussing the constraints of the cross-sectional study design and suggesting directions for future research.

Section 6 covers the cross-sectional constraint at length. It reads the results as associations rather than causal sequences, uses "predicts" and "is associated with" deliberately, and points to panel and longitudinal designs and to post-adoption outcomes as future work. In revision we widened the limitations further to take in level of analysis, common method variance, and the boundary conditions on generalizability.

Finally, carefully review the manuscript for minor grammatical, formatting, and citation inconsistencies.

The manuscript has been proofread for grammar and formatting, and the references and in-text citations were reconciled with our co-author's reference manager.

Do consider the following references...

We appreciate the suggestions. Having reviewed them, and following the Editor-in-Chief's note that recommended references may be set aside, we have kept our current citations, which we believe support the manuscript's arguments.

We thank the reviewers again for comments that have improved the paper. We hope the revision meets the journal's standard and look forward to your decision.

Yours sincerely,

The Authors

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Title: Examining AI–Blockchain Adoption in Life Insurance through an Integrated HOT-TOE Framework

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Examining AI-Blockchain Adoption in Life Insurance through an Integrated HOT-TOE Framework

Abstract

This study investigates the drivers and barriers influencing the Actual Adoption of Artificial Intelligence (AsI) and Blockchain technologies in the life insurance sector of Bangladesh. Integrating the Human–Organization–Technology (HOT-fit) and Technology–Organization–Environment (TOE) frameworks, a comprehensive conceptual model was developed to explain adoption behavior. Data were collected through a deductive questionnaire survey from 394 employees across 32 life insurance companies covering all eight divisions of Bangladesh using cluster sampling. The proposed hypotheses were examined using Partial Least Squares Structural Equation Modeling (PLS-SEM). The findings reveal that Organizational Readiness is the strongest predictor of Actual Adoption, primarily driven by Technology Readiness and Business Model Readiness. The positive association between readiness and adoption is strengthened by Perceived Cost considerations and attenuated by Stickiness to Legacy Systems. Notably, system integrity did not significantly predict readiness, suggesting that in compliance-driven markets security functions as an assumed regulatory baseline rather than a distinguishing driver of adoption. This study contributes to the literature by integrating HOT-fit and TOE in the context of AI-Blockchain adoption within life insurance in a developing economy and introduces novel constructs such as Stickiness to Legacy Systems and sustainability-oriented interoperability and cost-benefit perspectives.

Keywords Artificial Intelligence, Blockchain, Sustainability, HOT-fit, TOE, Life Insurance, Bangladesh

1 Introduction

1.1 Digital Transformation in Insurance

The rapid digital transformation in the global insurance sector driven by artificial intelligence (AI) and blockchain presents both opportunities and challenges for life insurance companies especially in emerging economies like Bangladesh. These technologies promise enhanced operational efficiency, fraud reduction, and improved customer experience, but their adoption is shaped by complex technological, organizational, human, and sustainability factors. Understanding these drivers and barriers is crucial for ensuring successful technological integration and sustainable competitive advantage. The insurance industry is increasingly leveraging AI and blockchain to streamline operations, improve data security, and foster innovation [1–3]. For example, AI can

automate claims processing, while blockchain ensures data integrity and transparency [4]. However, adoption rates remain uneven, particularly in developing countries, due to organizational readiness, regulatory challenges, and legacy system inertia [3,5]. This study examines the interplay of technology, human, organizational, and sustainability factors using combined HOT-fit and TOE frameworks to identify enablers and barriers to AI and blockchain adoption in life insurance, with a focus on Bangladesh as a representative emerging market [5,6].

Previous research has applied the TOE framework to analyze technology adoption in insurance and related sectors, highlighting the roles of technological advantage, organizational support, and environmental pressures. Studies have also shown that knowledge management and top management support are critical for successful adoption [2,3,6,7]. However, most research has focused on developed economies or specific industries, with limited empirical evidence from the Bangladeshi insurance context [5]. Furthermore, the integration of sustainability factors and the moderating effect of perceived cost remain underexplored [5,8,9]. Only a few studies have integrated sustainability factors and examined the moderating role of perceived cost in technology adoption models. Most existing studies focus on developed markets, leaving a gap in understanding the unique challenges and opportunities in emerging economies [5,8].

1.2 AI-Blockchain Adoption Scope in Insurance

A clarification on scope is warranted. While AI and blockchain are distinct technologies with different architectural properties and adoption maturity curves, this study examines them as a co-adopted technology bundle rather than separately. This treatment reflects the practical reality of InsurTech deployment, where AI-blockchain integration is increasingly implemented as a unified solution: AI provides the predictive intelligence and automation layer, while blockchain provides the immutable audit trail and smart contract execution layer [10]. Bibliometric evidence shows that the two technologies appear together in the majority of recent InsurTech adoption studies [10], and insurance companies in the sample reported evaluating and piloting them in tandem. Studying them as a bundle also reflects how organizational readiness is actually assessed: decision-makers evaluate their readiness for a digital transformation program, not for two independent technology procurement decisions. That said, we acknowledge that future research should disentangle the separate adoption drivers and barriers for each technology to provide more granular guidance. AI and blockchain technologies both play a significant role in reducing human error and automating processes, which enhances organizational security through improved fraud detection and secure data management. In life and annuity insurance, AI supports predictive mortality modeling and customer analytics, and blockchain enhances transparency in policy administration and actuarial processes [11]. Bibliometric and systematic reviews show rising research and InsurTech activity, especially on AI-driven risk models combined with blockchain-based smart contracts and peer-to-peer or parametric products [12]. Yet adoption remains uneven, with most implementations at prototype or limited-production stage, constrained by regulation, legacy systems, and unclear business cases [13]. Overall, the scope spans efficiency gains, new product designs, and trust-enhancing cyber-resilience, but requires careful governance and change management. In

response to the accelerating digital transformation within the life insurance industry and the limited empirical understanding of the determinants influencing AI and Blockchain adoption in developing economies, this study seeks to systematically examine the human, technological, organizational, and sustainability factors shaping Organizational Readiness and Actual Adoption through the following research questions:

RQ1: How do technological, human, organizational, and sustainability factors, examined through an integrated HOT-fit and TOE framework, shape Organizational Readiness and Actual Adoption of AI and Blockchain in life insurance?

RQ2: How does Organizational Readiness, within the integrated HOT-fit and TOE framework, serve as an enabler or barrier to AI and Blockchain implementation in the life insurance industry?

RQ3: How can the strategic implementation of AI and Blockchain in life insurance, guided by the HOT-fit and TOE frameworks, enhance Organizational Readiness and Actual Adoption, given the unique challenges and opportunities present in an emerging market context?

2 Literature Review

2.1 Theoretical Underpinning

This study integrates the Human-Organization-Technology fit (HOT-fit) model with the Technology-Organization-Environment (TOE) framework to explain AI and Blockchain adoption in the life insurance sector. Neither framework alone is sufficient for this phenomenon. Research relying solely on TOE captures organizational and environmental readiness well but consistently underspecifies the human dimension, the role of digital skills, resistance behaviors, and individual perceptions that determine whether technology is actually used after procurement [14]. Studies applying HOT-fit alone handle human and organizational fit well but lack systematic treatment of environmental pressures such as competitive dynamics, sustainability considerations, and interoperability with broader industry ecosystems. The integration bridges this gap deliberately: HOT-fit contributes the human and organizational fit lens, while TOE contributes the environmental and sustainability dimensions. Each construct maps onto one of the four combined-framework dimensions (Technology, Human, Organization, Sustainability), as shown in Figure 1. The HOT-fit model [15] emphasizes that system success is achieved when human, technological, and organizational dimensions are aligned. In the context of AI and blockchain adoption in life insurance, this is reflected in human factors such as digital literacy of blockchain and Stickiness to Legacy System, technological attributes such as system integrity, Technology readiness, and organizational aspects such as management support, and business model readiness [16] [17]. Organizational factors including management support and business model readiness reflect strategic alignment and internal capacity, while technological elements such as technology readiness and system integrity represent infrastructure maturity and perceived reliability [18]. Complementing this, the TOE framework expands the analysis by incorporating environmental and contextual pressures that influence adoption decisions. Interoperability and Perceived Organizational Benefits are conceptualized under the sustainability dimension, a deliberate

theoretical choice. Both constructs reflect long-term systemic value rather than short-term operational gain: interoperability determines whether AI-blockchain systems can be maintained and extended across organizational boundaries over time, and perceived organizational benefit captures the long-horizon investment logic that justifies adoption. This framing aligns with sustainability as a strategic driver of adoption rather than merely an environmental outcome [19] [20]. Prior research shows that blockchain-enabled integration enhances transparency, reduces redundancy, and optimizes resource utilization, thereby strengthening economic and environmental sustainability [19]. Similarly, perceived organizational benefits such as cost efficiency, improved competitiveness, and operational transparency reinforce sustainable performance outcomes. In the insurance context, AI-blockchain interoperability facilitates seamless inter-organizational data exchange, while relative advantage justifies continued investment, positioning sustainability not merely as an outcome but as a strategic driver of adoption intention [20]. Perceived cost operates as a critical boundary condition shaping the readiness–adoption relationship. By integrating HOT-fit and TOE, this research develops a multidimensional model where organizational readiness functions as a central mediating mechanism linking internal capabilities and contextual drivers to actual adoption. This integration is adopted as a deliberate theoretical choice suited to the multidimensional nature of AI-blockchain adoption; the study does not claim to empirically compare competing frameworks, but rather to demonstrate the explanatory value of their integration in this context.

2.2 Enablers and Barriers of AI-Blockchain Adoption in Life Insurance

Life and annuity (L&A) insurance is undergoing a broader InsurTech-driven digital transformation, in which AI, big data and blockchain are central technologies [21]. AI is used for predictive analytics, underwriting, claims triage, lapse and fraud prediction, and personalized product design, while blockchain supports shared ledgers, smart contracts and auditability across insurers, intermediaries and reinsurers [22]. Although many studies are cross-line, the underlying enablers and barriers they identify are directly relevant to life insurance, which is particularly data-intensive, long-tailed and regulation-heavy.

A dominant enabler is the potential for efficiency gains in underwriting, policy servicing and claims. AI-powered automation and machine-learning models reduce manual processing, improve risk selection and shorten cycle times [23]. Blockchain offers a single source of truth and tamper-resistant records, reducing reconciliation costs, disputes, and back-office overhead, especially when smart contracts automate premium collection, benefit triggers and payouts [24]. For life insurers facing margin pressure and legacy IT, these expected savings are a major adoption driver. AI enables more granular and dynamic risk assessment, incorporating heterogeneous data sources (medical records, wearables, behavioral data) to refine mortality and morbidity forecasts and personalize premiums and coverage [25]. Blockchain facilitates programmable policies and peer-to-peer or consortium-based risk sharing, opening space for novel life insurance designs and embedded products [26]. Bibliometric mapping shows risk assessment, fraud prevention and smart-contract–based insurance as core clusters in AI–blockchain insurance research [27]. Life

insurance involves long-term commitments and sensitive health and financial data. Studies identify trust and transparency as powerful enablers: blockchain's cryptographic immutability and distributed validation can increase confidence in record integrity, reduce disputes over beneficiary status or policy changes, and support regulator access to real-time audit trails [4]. At the organizational level, leadership commitment, strategic vision, and change management capacity are crucial. Interdisciplinary collaboration between actuarial, IT, and business teams, and in some cases, consortia and partnerships with InsurTechs and regulators also appears as a facilitator of blockchain consortia and shared platforms [12][27][24].

Across studies, regulatory complexity is one of the most persistent barriers. Life insurance is tightly regulated, with requirements on solvency, fair pricing, consumer protection, and data privacy. The opaque nature of some AI models raises concerns about explainability and non-discrimination in underwriting and claims decisions [28]. Blockchain introduces questions around data localization, GDPR-style erasure rights on immutable ledgers, legal status of smart contracts, and supervision of decentralized infrastructures [29]. Integration with legacy IT architectures is another central barrier. Empirical work on AI-blockchain cybersecurity in insurance points to difficulties connecting new platforms to old policy administration, claims and data warehouses, along with scalability constraints and inflexible architectures [4]. Systematic blockchain reviews emphasize that many insurers operate fragmented systems and siloed data, making end-to-end digital process redesign complex and costly [29]. AI's effectiveness depends on high-quality, well-governed data, yet studies highlight data quality problems, incomplete datasets, and governance weaknesses that can lead to biased risk assessments and governance risks [25]. For life insurance, sensitive health and genetic information amplify concerns around consent, secondary use and re-identification. Blockchain's transparency and permanence can conflict with privacy rules, and industry-wide data sharing faces cultural resistance and competitive concerns [30]. Reviews of blockchain in insurance stress unresolved issues around permissioned vs. public chains, access control, and key management for confidential life-insurance data [13]. Risk-averse cultures and inertia in product design further slow the shift toward highly personalized or connected-device-based life insurance, even where the technical potential exists [31]. From an economic perspective, high up-front investment, uncertain ROI, and the still-nascent maturity of some blockchain platforms discourage large-scale deployments [22]. Systematic analyses classify barriers along social, technological, environmental, economic and political (STEEP) dimensions, highlighting technological issues such as scalability, interoperability, latency, and energy consumption as ongoing concerns [11]. Across the literature, there is strong consensus that AI and blockchain can enable more efficient, transparent, and customer-centric life insurance through better risk modeling, automated smart contracts, and secure data sharing. However, implementation is constrained by regulatory uncertainty, legacy infrastructures, data governance issues, organizational resistance, and skills shortages. Three gaps in this literature motivate the present study. First, no existing study has combined HOT-fit and TOE to jointly examine the human, organizational, technological, and sustainability dimensions of AI-blockchain adoption in insurance, and the separate frameworks each leave blind spots that an integrated approach can

address. Second, empirical evidence from Bangladeshi and comparable South Asian emerging market insurance contexts is almost entirely absent; most studies draw on Western or East Asian samples, limiting generalizability to resource-constrained, regulation-light environments. Third, the moderating role of perceived cost in the readiness-to-adoption pathway has received only limited empirical attention in insurance technology research, despite cost being consistently identified as a dominant adoption barrier in practitioner surveys. This study addresses all three gaps.

2.3 Hypothesis Development

2.3.1 Business Model Readiness (BMR)

Business Model Readiness is identified as a significant organizational factor influencing the decision to adopt blockchain technology [32]. Organizational readiness is tied to the internal organizational aspect, suggesting that strategic intent and a proper strategic orientation are necessary preconditions for embracing innovative technologies [33]. Successful AI transformations require a solid business case that aligns with existing corporate strategies, which is critical for driving organizational change and enhancing readiness[34]. For organizations adopting solutions like blockchain, aligning the technology with a suitable and high-impact business use case is necessary to justify the required investment[35]. Therefore, we hypothesize that:

H1: Business model readiness positively influences the Organizational Readiness for adoption of AI-Blockchain in Insurance Industry.

2.3.2 Digital Literacy of Blockchain (DLB)

Organizational readiness is formally defined as the availability of required organizational resources for using an innovation, which specifically includes having employees with the necessary skills [32]. Digital literacy, referring to the capacity to use and interact with features in information technology devices, directly contributes to an individual's preparedness for adopting new technologies, such as blockchain [36]. Consequently, acquiring adequate human resource skills, encompassing employees' knowledge, experience, and technical background, is essential for organizations seeking effective development and utilization of technology like blockchain [37]. Studies on blockchain adoption confirm that the presence of sufficient organizational readiness, particularly the availability of employee resources and IT infrastructure, has a positive influence on a company's decision to adopt blockchain technologies [38]. Therefore, enhancing the digital literacy of organizational members concerning blockchain is a fundamental step toward building the requisite core competencies needed for technology implementation and ultimately enhancing organizational readiness [39]. Therefore, the study proposes the following hypothesis:

H2: Digital Literacy of Blockchain positively influences the Organizational Readiness for adoption of AI-Blockchain in Insurance Industry.

2.3.3 Interoperability (IO)

Greater interoperability with existing systems and standards is a significant technological factor influencing the successful adoption of new technologies [40]. Interoperability is defined as the degree to which an innovation aligns with the organization's existing values, previous experiences, and needs [41]. When systems lack interoperability or flexibility, the ability to be integrated and customized, it can present a barrier to seamless healthcare delivery and expedite the adoption of new systems [42]. Accordingly, organizations tend to adopt new technologies, like Green Information Technology (GIT), when they perceive them to be compatible with their existence [43]. Therefore, greater interoperability ensures that new systems can effectively work with the existing IT infrastructure, reducing the risk of extensive modifications and promoting a smoother transition, thereby increasing organizational readiness [44].

H3: Interoperability of technological systems positively influences the Organizational Readiness for adoption of AI-Blockchain in Insurance Industry.

2.3.4 Management Support (MS)

Stronger top management support is repeatedly identified as a critical organizational factor necessary for the successful adoption of new IT innovations like AI and blockchain [34]. Management support, defined as the degree to which executives understand and value the functionality of new technologies, ensures the allocation of necessary resources, including financial and organizational assistance, which is indispensable for applications like Generative AI [45]. Furthermore, top management commitment (leadership support) and resource provision are recognized as significant enablers of blockchain adoption and organizational readiness [32]. In the context of AI, management support positively influences the performance of adoption and is one of the strongest determinants, second only to competitive pressure in importance [45]. Therefore, strong management backing is vital because it drives organizational change, facilitates a supportive environment, and provides the strategic direction and resources required to enhance an organization's overall capability and willingness (readiness) to adopt these complex technologies [45]. which led to testing the following hypothesis:

H4: Management Support positively influences Organizational Readiness for adoption of AI-Blockchain in Insurance Industry.

2.3.5 Organizational Readiness (OR)

Organizational Readiness (OR) encompasses the availability of concrete and abstract resources, such as technical IT expertise, capabilities, and management systems, that companies possess to facilitate the implementation of new technology [46]. Sufficient organizational readiness, specifically concerning financial, employee, and IT infrastructure resources, has been consistently

identified as a significant enabler that positively influences a company's decision to adopt blockchain technologies [32]. Furthermore, conceptual models specifically suggest that organizational readiness (OR) positively affects the adoption of Generative AI (GEN-AI) among insurance companies [46]. Empirical evidence confirms this link, showing a significant positive relationship between organizational readiness and the adoption of Big Data Analytics (BDA), a capability that precedes and supports AI adoption [47]. Therefore, as organizational readiness reflects a firm's capability and preparedness, a higher level of this factor increases the likelihood of successfully implementing and achieving the actual adoption of AI and blockchain technologies [46]. Reviewing literature on the context of organizational readiness, some studies report inconsistent findings regarding the direct impact of OR on business outcomes. For example, while OR often predicts positive change, some research finds that its effect is context-dependent or mediated by other factors such as leadership or innovation climate [48,49]. In context of a study on sustainable economic performance, OR to learn was not always a moderator, suggesting that other variables may influence the relationship [49].

H5: Organizational Readiness positively influences to Actual Adoption of AI and blockchain in insurance industry.

2.3.6 Perceived Cost (PC)

Perceived cost plays a crucial role in shaping the relationship between organizational readiness and actual adoption of new technologies. When organizations perceive high implementation costs, even those with strong readiness may hesitate to proceed with adoption. Empirical studies confirm that perceived cost significantly moderates the effect of readiness on adoption intentions, especially for blockchain. High perceived costs can diminish the positive impact of organizational readiness, while lower costs can enhance the likelihood of actual adoption. Importance-Performance Map Analysis further highlights perceived cost as a critical factor influencing adoption decisions, often outweighing other enablers; some studies report that perceived cost does not always have a straightforward moderating effect. For example, in a study on blockchain adoption, perceived cost was not a mediator but a moderator, and its effect varied depending on other enabling factors [50,51]. In another study, perceived cost was found to negatively impact behavioral intentions, but in some contexts, organizations with high readiness still proceeded with adoption despite high costs [52,53]. Therefore, managing perceived cost is essential for organizations aiming to translate readiness into successful technology adoption [50,51,54].

H6: Perceived Cost moderates the relationship between Organizational Readiness and Actual Adoption.

2.3.7 Perceived Organizational Benefit (POB)

The degree to which a firm perceives an invention as superior to its prior version, known as Perceived Organizational Benefits, has been consistently identified as an essential technological predictor for innovation adoption and usage [41]. Specifically, regarding AI, achieving a higher

level of Perceived Benefits through faster business activities and increased operational efficiency is recognized as a key element influencing its adoption by insurance companies [46]. Understanding the intrinsic value and benefits of adopting technologies like AI is considered a fundamental prerequisite for successful implementation and enhancing organizational readiness [55]. For complex innovations like blockchain, firms whose participants perceive the innovation as more beneficial are more likely to implement them thoroughly, suggesting perceived value increases adoption likelihood [32]. This alignment of perceived value to expected outcomes is necessary because organizations rely on technological advantages to justify the transformation decision and enhance their capabilities and willingness for adoption [40].

H7: Perceived Organizational Benefit positively influences Organizational Readiness for adoption of AI-Blockchain in Insurance Industry.

2.3.8 System Integrity (SI)

Blockchain inherently provides high system integrity through its use of cryptographic techniques and a distributed, shared, and immutable ledger, which is poised to revolutionize trusted transactions [50]. For AI technology, high system quality is recognized as a key factor affecting operational efficiency and organizational performance because a good system can better meet user requirements [45]. Furthermore, combining AI with blockchain allows for automated claim verification and detection of fraudulent activity using tamper-proof records, thereby reducing uncertainty and building confidence [56]. This technological assurance is critical because reliability and information security have been empirically shown to have significant positive effects on an organization's belief and trust in using new IT services [33]. Organizational readiness, encompassing technical IT expertise and infrastructure, is highly dependent on the availability of sufficient resources needed to manage and invest in the adoption of new technology, which is ultimately enabled by this technological confidence [47]. Thus, evidence supports the following hypothesis:

H8: System Integrity positively influences Organizational Readiness for adoption of AI-Blockchain in Insurance Industry.

2.3.9 Stickiness to Legacy System (SLS)

Legacy systems are a significant barrier to digital transformation, as their complexity and integration challenges hinder organizational change and innovation [57,58]. Organizations often maintain legacy systems to avoid operational disruptions, but this results in inefficiencies and missed opportunities for transformation [57,58]. The persistence of outdated technology slows down the adoption of new digital tools and processes, directly impacting organizational readiness for innovation. Risk-averse cultures and reluctance to overhaul existing systems further exacerbate this stickiness, making it difficult for organizations to adapt to technological advancements [57,58]. In the insurance sector, integration with legacy systems is frequently cited as a major

obstacle to adopting AI and blockchain technologies, limiting the organization's ability to respond to market changes and regulatory requirements [4,59]. Hence:

H9: Stickiness to Legacy System negatively influences Organizational Readiness for adoption of AI-Blockchain in Insurance Industry.

2.3.10 Technology Readiness (TR)

Technology readiness is a critical enabler for organizations aiming to adopt advanced digital solutions and drive transformation [60,61]. Higher levels of technology readiness equip organizations with the necessary infrastructure, skills, and mindset to integrate new technologies effectively [61,62]. Empirical studies show that technology readiness directly improves employee capacity building, which is essential for organizational preparedness and adaptability [61]. Organizations with robust technology readiness are better positioned to respond to digital disruption and seize emerging opportunities [60,63]. Research also highlights that technological resources, such as IT infrastructure and digital capabilities, are foundational dimensions of organizational readiness [60,62]. Therefore, the following hypothesis is developed:

H10: Technology Readiness positively influences Organizational Readiness for adoption of AI-Blockchain in Insurance Industry.

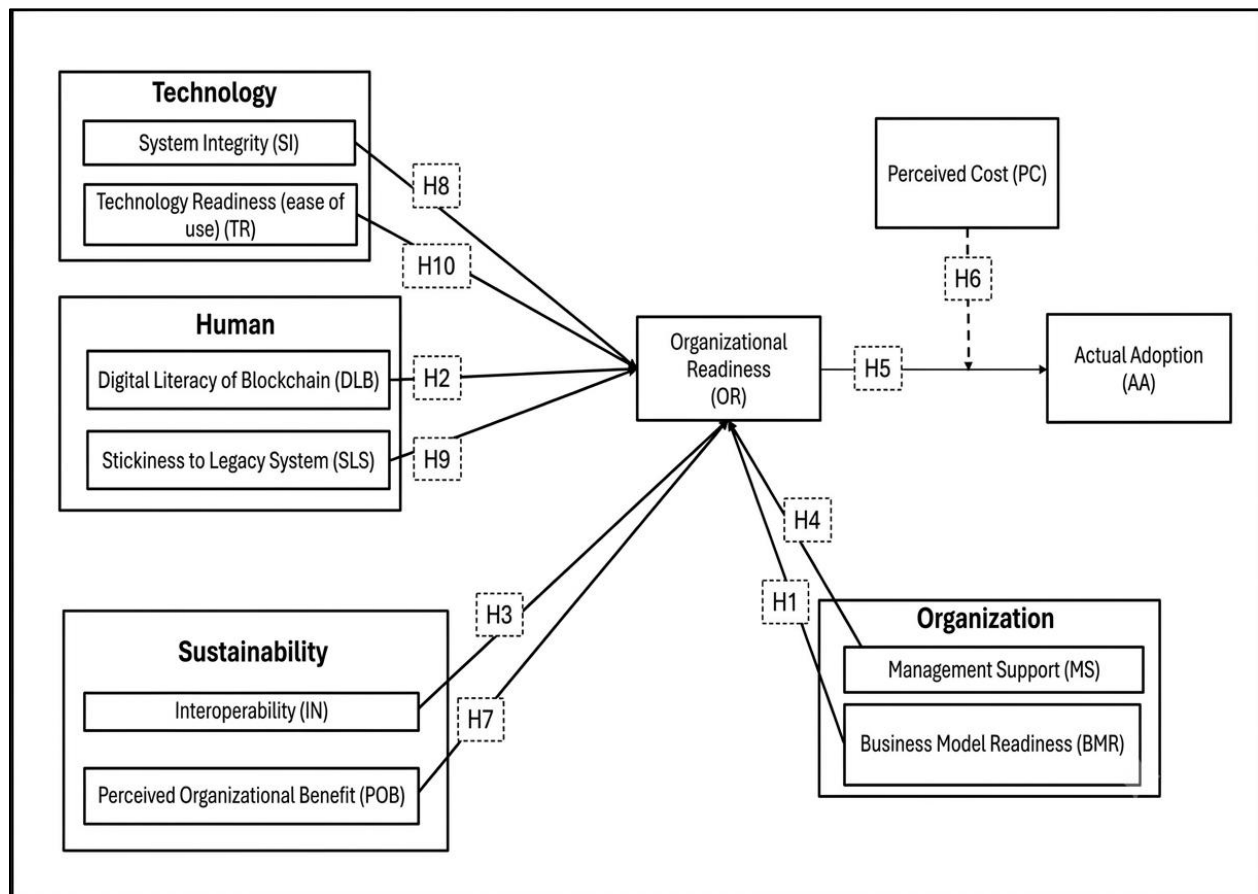


Figure 1: Conceptual Framework

3 Methodology

The study employed a two-stage cluster sampling design. Cluster sampling is appropriate when the population is organized into identifiable subgroups and a comprehensive individual-level frame is unavailable [64]. A complete individual-level sampling frame of insurance employees is not publicly available in Bangladesh, which precludes simple random sampling; cluster sampling by firm and role is therefore both a practical necessity and a means of ensuring coverage across ownership types, divisions, and functional roles. In the first stage, life insurance companies registered with the Insurance Development and Regulatory Authority (IDRA) of Bangladesh were grouped by ownership type (foreign, public, and private), and 32 firms were selected across all eight administrative divisions. In the second stage, employees were recruited from four core operational roles within each firm: Underwriting, Claims Department, Customer Service and Conservation, and Accounting and Finance. A minimum of four responses per role per firm was required to ensure functional representation and reduce within-firm bias. Employee-level data collection for organizational constructs is consistent with established practice in technology adoption research. This study measures organizational constructs, such as Organizational Readiness, Business Model Readiness, and Management Support, through individual employee perceptions. The unit of analysis is the individual respondent, and PLS-SEM was estimated at the individual level ($N = 394$). Consistent with established practice in perception-based technology adoption research, these constructs capture employees' perceptions of organizational conditions rather than objective, firm-level states measured through aggregated or archival data [65]. The multi-role sampling design, drawing respondents from underwriting, claims, customer service, and finance functions within each firm, was intended to capture a broad cross-section of perceptions rather than to construct firm-level aggregate scores. A structured electronic questionnaire was distributed via Microsoft Forms between August 11 and October 29, 2025, yielding 456 initial responses. After 62 redundant submissions were excluded, the final sample was 394. G*Power calculations indicated a minimum of 68 to 160 respondents [66,67], thus ensuring strong statistical power. Responses were collected from all eight administrative divisions of Bangladesh, with major contributions from insurance firms.

A six-point Likert-type scale (1 = “strongly disagree” and 6 = “strongly agree”) was chosen over more common five- or seven-point formats. The six-point scale removes the neutral midpoint, which prevents respondents from defaulting to a non-committal response, a central tendency bias that is especially prevalent in organizational surveys where social desirability effects operate [68,69]. Removing this option produces more discriminating response distributions and has been empirically validated in technology adoption research [68]. The survey instrument, administered through Microsoft Forms, served as the empirical basis for testing and validating the proposed conceptual model and associated hypotheses. All construct items were adapted from validated

instruments in prior technology adoption studies. A pilot test was conducted with 22 insurance professionals (excluded from the final sample) to assess item clarity, face validity, and completion time. Based on pilot feedback, minor wording adjustments were made to three items in the Digital Literacy of Blockchain and Stickiness to Legacy Systems constructs. Content validity was confirmed through a panel of three insurance industry experts and two academic experts in technology adoption, whose feedback shaped the final instrument. Data were then analyzed using SmartPLS 4, employing Partial Least Squares Structural Equation Modeling (PLS-SEM). Partial Least Squares Structural Equation Modeling (PLS-SEM) was selected over covariance-based SEM for several reasons aligned with the study's aims. First, the model is complex, comprising eleven constructs, forty-seven indicators, and a moderation term; PLS-SEM accommodates such complexity without the identification and convergence problems that can affect covariance-based estimation. Second, the study is prediction-oriented, its central concern is explaining variance in Organizational Readiness and Actual Adoption, which aligns with PLS-SEM's focus on maximizing explained variance in target constructs. Third, PLS-SEM estimates interaction effects, such as the moderating role of Perceived Cost, in a straightforward manner. These characteristics make PLS-SEM the appropriate estimator for the present model. All analyses were conducted at the individual respondent level (N = 394); responses were not aggregated to the firm level. The primary measurement and structural model was estimated in SmartPLS 4. Supplementary discriminant-validity analyses reported in Section 4.1.2, comprising bias-corrected bootstrap confidence intervals for the HTMT ratios (5,000 resamples) and a confirmatory factor analysis comparing a two-factor and a one-factor specification of Organizational Readiness and Actual Adoption, were conducted in R (version 4.x) using the *semnr* and *lavaan* packages to enable open, reproducible replication.

4. Results and Analysis

Variable	Category	Frequency (n = 394)	Percentage (%)
Gender	Male	242	61.4%
	Female	152	38.6%
Age Group	18–22 years	48	12.2%
	23–27 years	183	46.4%
	28–32 years	109	27.7%
	33–37 years	54	13.7%
Education Level	Bachelor's Degree	287	72.8%
	Master's Degree	107	27.2%
Job Level	Entry-Level	174	44.2%

	Mid-Level	148	37.6%
	Senior-Level / Manager	72	18.3%
Years of Experience	Less than 1 years	52	13.2%
	1–3 years	176	44.7%
	4–6 years	103	26.1%
	Above 6 years	63	16.0%
Department / Division	Underwriting	122	31.0%
	Claims Department	98	24.9%
	Customer Service & Conservation	114	28.9%
	Accounting and Finance	60	15.2%

Table 1: Demographic profile of the respondents

4.1 Measurement Model Assessment

4.1.1 Reliability and Validity Test

Internal consistency reliability is largely satisfactory, as most constructs have Cronbach's alpha and Composite reliability (ρ_a) values above the conventional threshold of 0.70 [70]. For example, SLS has a very high ρ_a of 0.906, and AA is strong with a ρ_a of 0.845. However, Management Support (MS) shows a Cronbach's alpha of 0.613, which falls below the 0.70 threshold [70]. Cronbach's alpha assumes tau-equivalence among indicators, meaning equal item loadings, an assumption rarely met in practice that systematically underestimates reliability in PLS-SEM [71]. The more appropriate reliability metric in variance-based SEM is composite reliability (ρ_a is 0.757, exceeding the 0.70 benchmark. The AVE for MS is 0.707, the highest across all constructs, confirming strong convergent validity. MS also has only two indicators; alpha is sensitive to scale length and underestimates reliability for short scales. Composite reliability and AVE together justify retaining this construct.

Convergent validity is confirmed for all constructs, as every Average Variance Extracted (AVE) value is well above the required threshold of 0.50 [71]. This signifies that more than 50% of the variance in the indicators is explained by their respective construct. For example, Perceived Cost (PC) has the highest AVE of 0.785, meaning approximately 78.5% of its indicator variance is attributable to the PC construct. The lowest AVE is 0.568 (TR), which is still acceptable. Overall,

despite the low Cronbach's alpha for MS, the high Composite Reliability and AVE across the board indicate that the items consistently and validly measure the intended constructs.

Constructs	Cronbach's alpha	Composite reliability (ρ_a)	Average variance extracted (AVE)
AA	0.843	0.845	0.615
BMR	0.828	0.828	0.592
DLB	0.811	0.812	0.570
IN	0.799	0.801	0.624
MS	0.613	0.757	0.707
OR	0.819	0.820	0.581
PC	0.726	0.726	0.785
POB	0.817	0.820	0.578
SI	0.763	0.766	0.678
SLS	0.835	0.906	0.591
TR	0.810	0.810	0.568

Table 2: Measurement Model

4.1.2 Discriminant Validity Test

The Heterotrait-Monotrait Ratio (HTMT) Matrix generally confirms discriminant validity [72], as most values are significantly below the conservative threshold of 0.85, or the more liberal 0.90 [73]. The values represent the ratio of average correlations between indicators of different constructs (heterotrait) to the average correlations of indicators within the same construct (monotrait). For example, the HTMT ratio between Management Support (MS) and Stickiness to Legacy Systems (SLS) is 0.745, confirming they are clearly distinct constructs. Two HTMT ratios exceed the conservative 0.90 threshold: OR–AA (0.945) and TR–OR (0.920). To assess whether these reflect a genuine lack of discriminant validity or the expected proximity of theoretically adjacent constructs, two tests were conducted. First, bias-corrected bootstrap confidence intervals (5,000 resamples) were computed for both ratios. The HTMT inference criterion requires that the interval exclude 1.00 [74] Both satisfy this: OR–AA yields a 95% CI of [0.886, 0.994] and TR–OR yields [0.854, 0.981], with neither upper bound reaching unity. Second, a confirmatory factor analysis compared a two-factor model, specifying OR and AA as separate correlated constructs, against a one-factor model collapsing all ten indicators onto a single dimension. The two-factor model fit significantly better ($\Delta\chi^2 = 11.3$, $\Delta df = 1$, $p < 0.001$), and its fit indices were acceptable ($\chi^2 = 72.1$, $df = 34$, $RMSEA = 0.053$, $CFI = 0.978$) and superior to the one-factor solution ($\chi^2 = 83.4$, $df = 35$, $RMSEA = 0.059$). This indicates that OR and AA are empirically separable rather than redundant. The estimated inter-factor correlation (0.943) confirms that the two constructs, while distinct, are strongly related, consistent with readiness functioning as the proximal antecedent of adoption. Together, the confidence-interval and model-comparison evidence supports the discriminant validity of the OR–AA and TR–OR pairs, while acknowledging the high correlation inherent in theoretically adjacent readiness and adoption constructs.

Constructs	AA	BMR	DLB	IN	MS	OR	PC	POB	SI	SLS	TR	PC x OR
AA												
BMR	0.864											
DLB	0.777	0.796										
IN	0.857	0.896	0.833									
MS	0.112	0.071	0.067	0.084								
OR	0.945	0.888	0.814	0.893	0.137							
PC	0.779	0.779	0.806	0.843	0.084	0.839						
POB	0.800	0.880	0.780	0.857	0.065	0.858	0.854					
SI	0.770	0.763	0.800	0.878	0.074	0.811	0.781	0.791				
SLS	0.137	0.119	0.153	0.092	0.745	0.122	0.089	0.068	0.066			
TR	0.855	0.893	0.807	0.891	0.084	0.920	0.811	0.849	0.875	0.092		
PC x OR	0.047	0.089	0.037	0.037	0.307	0.046	0.060	0.080	0.036	0.231	0.104	

Table 3: HTMT Matrix

	OR–AA	TR–OR
HTMT	0.945	0.920
95% CI (bias-corrected)	[0.886, 0.994]	[0.854, 0.981]

Table 3a: HTMT ratios with bias-corrected bootstrap confidence intervals

Model	Chi-sq (χ^2)	df	RMSEA	CFI
One-factor	83.4	35	0.059	—
Two-factor	72.1	34	0.053	0.978
$\Delta(1f - 2f)$	11.3	1	—	—

Table 3b: Confirmatory factor analysis, OR–AA

4.1.3 Indicator Loadings and Multicollinearity Test

Table 4 confirms the reliability of the measurement model and the absence of significant multicollinearity issues. Outer Loadings are all strong, exceeding the minimum threshold of 0.708 (or 0.70), which indicates that the constructs are well-represented by their respective indicators. Both Perceived Cost indicators load strongly (PC-1 = 0.884, PC-2 = 0.887), and the construct meets the reliability and validity thresholds for a two-indicator reflective construct (AVE = 0.786, composite reliability = 0.880). The second indicator, PC-2, loads at 0.884, confirming that the construct is well-measured. Two-item reflective constructs are recognized as acceptable in PLS-SEM when AVE exceeds 0.50 and composite reliability exceeds 0.70, both of which are met for

PC (AVE = 0.785, CR = 0.726). Regarding multicollinearity, all VIF (Variance Inflation Factor) values are well below the conservative threshold of 3.0 and the highly stringent 5.0 limit indicating the absence of pathological collinearity and no evidence of substantial common method bias. With the highest VIF being 1.929 for BMR-3. This is crucial as VIF measures how much the variance of an estimated regression coefficient is increased due to collinearity; values under 5.0 (and ideally under 3.0) [75] confirm that the exogenous constructs are sufficiently independent. To directly address common method bias risk arising from single-source, single-timepoint data collection, two post-hoc checks were conducted. First, Harman's single-factor test was performed by constraining all study variables to one factor in an exploratory factor analysis. The first unrotated factor explained 31.4% of the total variance, well below the 50% threshold indicating serious common method bias [76] Second, the full collinearity assessment [76] confirmed that all construct-level VIF values remain below 3.3. As a further check, and following the reviewer-recommended marker-variable logic, a marker-based assessment was conducted using the Lindell and Whitney (2001) approach. Management Support served as an ad-hoc marker, as it exhibited the lowest average correlation with the remaining constructs. The implied method-variance estimate was approximately 4%, and CMV-adjusted construct correlations differed from their unadjusted values by 0.013 or less. All substantive relationships retained their significance and magnitude after adjustment. Notably, two constructs measured through the same instrument, Management Support and Stickiness to Legacy Systems, correlate near zero with the remaining constructs, a pattern inconsistent with the systematic inflation that common method variance would produce. Together, these diagnostics indicate that common method bias is unlikely to materially distort the structural findings. Finally, the Excess Kurtosis and Skewness values are small and close to zero across all indicators suggest that the data distribution is relatively normal and free from severe non-normality concerns.

Constructs	Outer loadings	VIF
AA-1	0.803	1.858
AA-2	0.790	1.802
AA-3	0.801	1.849
AA-4	0.774	1.749
AA-5	0.753	1.619
BMR-1	0.773	1.761
BMR-2	0.743	1.530
BMR-3	0.810	1.929
BMR-4	0.770	1.701
BMR-5	0.752	1.618
DLB-1	0.727	1.470
DLB-2	0.780	1.802
DLB-3	0.716	1.552
DLB-4	0.750	1.719
DLB-5	0.798	1.797
IN-1	0.805	1.666
IN-2	0.809	1.713

IN-3	0.768	1.535
IN-4	0.777	1.565
MS-1	0.928	1.242
MS-2	0.743	1.242
OR-1	0.773	1.666
OR-2	0.748	1.568
OR-3	0.769	1.684
OR-4	0.793	1.743
OR-5	0.726	1.502
PC-1	0.884	1.480
PC-2	0.887	1.480
POB-1	0.763	1.645
POB-2	0.771	1.639
POB-3	0.778	1.699
POB-4	0.710	1.446
POB-5	0.777	1.624
SI-1	0.821	1.587
SI-2	0.827	1.480
SI-3	0.822	1.597
SLS-1	0.773	1.710
SLS-2	0.710	1.782
SLS-3	0.784	1.791
SLS-4	0.735	1.781
SLS-5	0.837	1.621
TR-1	0.758	1.628
TR-2	0.733	1.460
TR-3	0.795	1.821
TR-4	0.758	1.582
TR-5	0.723	1.536
PC x OR	1.000	1.000

Table 4: Indicator loadings and multicollinearity test

4.2 Structural Model Assessment

The structural model demonstrated strong predictive relevance, thereby empirically validating the integrated HOT-TOE framework. The predictive power in Table 6 was confirmed by the Stone-Geisser [77,78] Q^2 statistic for all indicators of the key endogenous constructs, Actual Adoption (AA) and Organizational Readiness (OR). For Actual Adoption, the Q^2 values ranged from 0.348 to 0.411, and for Organizational Readiness, they ranged from 0.358 to 0.443. Since all Q^2 values are substantially greater than zero, the model confirms that the structural paths derived from the combined framework accurately predict the variance in both the mediating and dependent constructs [79]

4.2.1 Result of structural model and hypothesis testing

The structural model results presented in Table 5 indicate that nine of the ten hypothesized paths are statistically supported. The exception is the relationship between System Integrity (SI) and Organizational Readiness (OR), which does not reach significance at $\alpha = 0.05$. Notably, a reporting error in the original submission has been corrected: the path from Stickiness to Legacy Systems (SLS) to Organizational Readiness (SLS $\rightarrow$ OR, $\beta = -0.114$, $p = 0.032$) is supported, while System Integrity (SI $\rightarrow$ OR, $\beta = 0.086$, $p = 0.100$) is not. For example, Business Model Readiness (BMR) has a positive and significant effect on Organizational Readiness ($\beta = 0.182$, $t = 2.831$, $p = 0.005$, $f^2 = 0.094$). According to Cohen's [56] guidelines ($f^2 \geq 0.02$ = small, ≥ 0.15 = medium, ≥ 0.35 = large), this represents a small effect size. The strongest relationship in the model is between Organizational Readiness and Actual Adoption ($\beta = 0.673$, $t = 17.009$, $p < 0.001$, $f^2 = 0.738$), indicating a large effect and underscoring the central role of readiness in driving adoption outcomes. Similarly, Technology Readiness significantly influences Organizational Readiness ($\beta = 0.247$, $t = 4.443$, $p < 0.001$, $f^2 = 0.167$), reflecting a medium effect and confirming that technological infrastructure is a substantial determinant of readiness. Perceived Organizational Benefit shows a small effect on Organizational Readiness ($\beta = 0.161$, $t = 2.472$, $p = 0.013$, $f^2 = 0.033$). The direct effect of Perceived Cost on Actual Adoption is also significant ($\beta = 0.180$, $t = 4.223$, $p < 0.001$, $f^2 = 0.153$), indicating that cost perceptions directly shape adoption beyond their moderating role. The interaction term (PC $\times$ OR $\rightarrow$ AA) is likewise significant ($\beta = 0.053$, $t = 2.548$, $p = 0.011$, $f^2 = 0.097$), confirming that Perceived Cost moderates the Organizational Readiness–Actual Adoption relationship. The interaction term was constructed using the product indicator approach recommended by Hair et al. [75] in SmartPLS 4, without mean-centering, consistent with the two-stage moderation procedure. In contrast, the effect of System Integrity on Organizational Readiness ($\beta = 0.086$, $t = 1.643$, $p = 0.100$, $f^2 = 0.010$) is not statistically significant and demonstrates a negligible effect. Overall, interpreting effect sizes alongside statistical significance provides a more comprehensive understanding of the practical relevance of each relationship within the model. To interpret the significant Perceived Cost effects, which comprise both a direct path ($\beta = 0.180$) and an interaction with Organizational Readiness ($\beta = 0.053$), a simple slopes analysis was conducted. The effect of Perceived Cost on Actual Adoption was positive and significant across all levels of Organizational Readiness, strengthening as readiness increased: at low readiness (-1 SD) the slope was 0.124 ($p = 0.006$), at mean readiness 0.178 ($p < 0.001$), and at high readiness ($+1$ SD) 0.232 ($p < 0.001$). A Johnson-Neyman analysis confirmed that the Perceived Cost effect remained significantly positive across effectively the entire observed range of Organizational Readiness. Thus the moderation does not reverse or nullify the Perceived Cost effect at any realistic level of readiness; rather, organizational readiness amplifies the influence of Perceived Cost on adoption.

Hypothesis	Constructs	Path Coefficients	Standard deviation	f-square	T-values	P-values	Decision
H1	BMR $\rightarrow$ OR	0.182	0.064	0.094	2.831	0.005	Supported

H2	DLB → OR	0.117	0.056	0.092	2.096	0.036	Supported
H3	IN → OR	0.149	0.067	0.094	2.235	0.025	Supported
H4	MS → OR	0.136	0.051	0.083	2.643	0.008	Supported
H5	OR → AA	0.673	0.040	0.738	17.009	0.000	Supported
H6	PC x OR → AA	0.053	0.021	0.097	2.548	0.000	Supported
H7	POB → OR	0.161	0.065	0.033	2.472	0.013	Supported
H8	SI → OR	0.086	0.052	0.010	1.643	0.100	Not Supported
H9	SLS → OR	-0.114	0.053	0.097	2.140	0.032	Supported
H10	TR → OR	0.247	0.056	0.167	4.443	0.000	Supported

Table 5: Hypothesis Testing

4.2.2 Model Explanatory and Predictive Power

The results for the Coefficient of Determination (R^2) and Predictive Relevance (Q^2) confirm the structural model's robust explanatory power and out-of-sample predictive capability. The R^2 values for the endogenous constructs are 0.699 for Organizational Readiness (OR) and 0.643 for Actual Adoption (AA). These values, particularly $R^2 = 0.699$, which is close to 0.70, are considered substantial [80], indicating that 69.9% of the variance in OR and 64.3% of the variance in AA are successfully explained by their antecedent constructs in the model. This is a strong finding for the model's in-sample fit. Furthermore, the Predictive Relevance (Q^2 predict) confirms the model's ability to predict new, unseen data. For a model to have predictive relevance, all Q^2 values must be greater than zero as stated by [81]. As shown, all indicators for both OR and AA exhibit Q^2 values significantly above zero (e.g., OR-4 is 0.443, and AA-1 is 0.411). This substantial positive value demonstrates that the structural model is able to accurately predict the data points of the endogenous constructs, supporting the generalizability and robustness of the model's findings beyond the immediate sample.

Constructs	R-square	Constructs	Q^2 predict
		AA-1	0.411
		AA-2	0.397
		AA-3	0.358
		AA-4	0.348
AA	0.643	AA-5	0.364
OR	0.699	OR-1	0.401
		OR-2	0.358
		OR-3	0.381
		OR-4	0.443
		OR-5	0.358

Table 6: Coefficient of Determination (R^2) & Predictive Relevance (Q^2)

4.2.3 Mediation Analysis: Indirect Effects via Organizational Readiness

To assess the mediating role of Organizational Readiness (OR) in the HOT-TOE framework, specific indirect effects were estimated using bootstrapping with 5,000 resamples and 95% bias-corrected confidence intervals, following established PLS-SEM practice [75] Table 7 presents the indirect effects of each antecedent on Actual Adoption (AA) through OR as the mediator. Seven of the eight indirect paths are statistically significant, as their bias-corrected confidence intervals do not include zero. Technology Readiness exerts the strongest indirect effect ($\beta = 0.167$, CI [0.092, 0.243]), followed by Business Model Readiness ($\beta = 0.122$, CI [0.036, 0.208]) and Perceived Organizational Benefit ($\beta = 0.109$, CI [0.026, 0.200]). Interoperability ($\beta = 0.101$, CI [0.012, 0.190]), Management Support ($\beta = 0.091$, CI [0.031, 0.158]), and Digital Literacy of Blockchain ($\beta = 0.079$, CI [0.005, 0.151]) also transmit significant positive indirect effects through Organizational Readiness. Stickiness to Legacy Systems shows a significant negative indirect effect ($\beta = -0.077$, CI [-0.143, -0.007]), confirming that legacy system inertia not only impedes readiness directly but also suppresses adoption indirectly through that pathway. The indirect effect of System Integrity on Actual Adoption through OR ($\beta = 0.058$, CI [-0.007, 0.130]) is not significant, consistent with the non-significant direct path reported in H8 and reinforcing the interpretation that system integrity functions as a baseline expectation rather than a readiness-building factor in this context. This interpretation is reinforced by the mediation results: system integrity exerts no significant indirect effect on adoption through readiness either, indicating that its non-significance is not an artifact of the direct path but a consistent pattern across the model.

Indirect Path	β	2.5% CI	97.5% CI	P-value	Decision
TR $\rightarrow$ OR $\rightarrow$ AA	0.167	0.092	0.243	0.000	Significant
BMR $\rightarrow$ OR $\rightarrow$ AA	0.122	0.036	0.208	0.005	Significant
POB $\rightarrow$ OR $\rightarrow$ AA	0.109	-0.143	-0.007	0.014	Significant
IN $\rightarrow$ OR $\rightarrow$ AA	0.101	0.012	0.190	0.028	Significant
MS $\rightarrow$ OR $\rightarrow$ AA	0.091	0.031	0.158	0.009	Significant
DLB $\rightarrow$ OR $\rightarrow$ AA	0.079	0.005	0.151	0.037	Significant
SLS $\rightarrow$ OR $\rightarrow$ AA	-0.077	-0.143	-0.007	0.034	Significant
SI $\rightarrow$ OR $\rightarrow$ AA	0.058	-0.007	0.130	0.104	Not Significant

Table 7: Specific Indirect Effects via Organizational Readiness (OR)

5. Discussion of Results and Implications

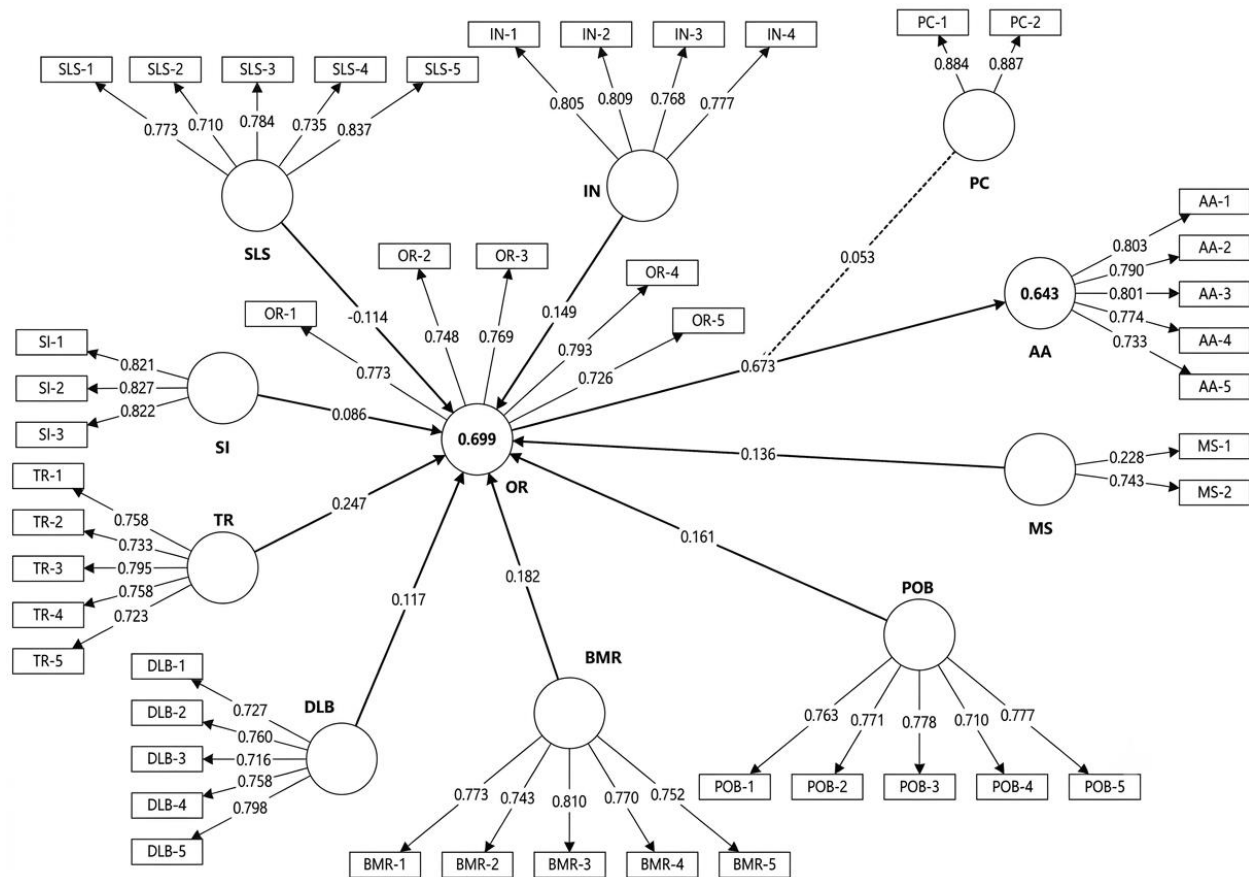


Figure 2: PLS-SEM Model

Before discussing the individual findings, it is useful to map the results back to the three research questions posed in Section 1.2. RQ1 asked how the integrated framework's factors shape readiness and adoption: the model proved empirically valid, with nine of ten paths supported and R^2 values of 0.699 (OR) and 0.643 (AA), demonstrating that the combined framework explains adoption well in this context. RQ2 asked how Organizational Readiness acts as an enabler or barrier: the findings confirm it as a powerful enabler ($\beta = 0.673$, $p < 0.001$), with readiness itself shaped by six significant antecedents and constrained by legacy system inertia. RQ3 asked how strategic AI-blockchain implementation guided by the framework can enhance readiness and adoption: the significant roles of Business Model Readiness, Management Support, and Perceived Organizational Benefit point to the specific strategic levers that organizations should activate. The structural model proved highly effective, with nine out of ten hypothesized paths showing statistical significance, validating the combined HOT-fit and TOE framework. Crucially, Organizational Readiness (OR) was the strongest predictor of Actual Adoption (AA) ($\beta = 0.673$, $p < 0.001$). The dominance of Organizational Readiness ($\beta = 0.673$) is consistent with prior readiness-adoption research, and its magnitude positions readiness as the single most decisive determinant in the model. This strong relationship confirms that employees' perceptions of their organization's internal health, including adequate infrastructure, committed management, and

competent staff, are the primary driver of their reported adoption of AI and blockchain. This powerful finding supports the core principle of the HOT-fit model: aligning human, organizational, and technological elements is essential for system success, mirroring results found in past technology adoption research [82].

Consistent with the TOE framework, Technology Readiness (TR) emerged as a critical enabler of Organizational Readiness (OR) ($\beta = 0.247$, $p < 0.001$). This significant relationship indicates that employees who perceive well-developed IT infrastructure and user-friendly systems also report greater organizational preparedness to implement new digital solutions [61]. The importance of internal organizational determinants was further reinforced by the significant effects of Business Model Readiness (BMR) on Organizational Readiness ($\beta = 0.182$, $p = 0.005$) and Management Support (MS) on Organizational Readiness ($\beta = 0.136$, $p = 0.008$). These results highlight that strategic alignment, leadership commitment, and resource allocation play essential roles in strengthening organizational capability and willingness to adopt AI and blockchain technologies, consistent with earlier empirical studies on technology adoption. [32].

The results confirm that readiness is driven by a variety of factors. On the human side, Digital Literacy of Blockchain (DLB) ($\beta = 0.117$, $p = 0.036$) significantly increased readiness, thereby validating the human component of the HOT model. Beyond personnel, key technological and environmental enablers also played a role: Interoperability (IN) ($\beta = 0.149$, $p = 0.025$) and Perceived Organizational Benefit (POB) ($\beta = 0.161$, $p = 0.013$). These factors boost readiness by promising tangible improvements in areas like operational efficiency, data integration, and trust. These findings are consistent with existing literature [40,46] that identifies the perception of technological advantages as a critical precursor to adoption.

Stickiness to Legacy Systems (SLS) showed a significant negative association with Organizational Readiness ($\beta = -0.114$, $p = 0.032$). This finding is noteworthy: even controlling for all positive readiness drivers, entrenched legacy systems pull in the opposite direction. Insurance companies built around legacy policy administration platforms face structural drag on their capacity to absorb new digital technologies, consistent with the broader financial services literature [4,57]. Furthermore, Perceived Cost (PC) significantly moderated the OR $\rightarrow$ AA relationship ($\beta = 0.053$, $p = 0.011$), implying that if the costs associated with adoption are too high, they can diminish the positive effects of organizational readiness [51]. This extends prior evidence that perceived cost moderates the readiness-adoption relationship for blockchain: our results show the interaction is amplifying rather than attenuating, with cost considerations intensifying the readiness-adoption link rather than simply dampening it. The one unsupported hypothesis was SI $\rightarrow$ OR ($\beta = 0.086$, $p = 0.100$). This is arguably the study's most interesting null result. System integrity, namely the cryptographic immutability and tamper-resistance that blockchain provides, is widely cited as a key enabler in the InsurTech literature. Its non-significance here is a substantive contextual finding: in the Bangladeshi insurance sector, data security appears to be perceived as a regulatory given rather than a readiness differentiator. Respondents working under IDRA compliance frameworks likely treat security as a minimum standard all systems must satisfy, not a feature that actively

builds preparedness. This context-specificity is meaningful for researchers applying HOT-fit in other regulated industries, and deviates from some prior findings [50]. This null result is the study's most theoretically consequential finding. System integrity, the cryptographic immutability and tamper-resistance that constitute blockchain's central value proposition, is almost universally treated in the InsurTech literature as a primary driver of adoption. Its non-significance here inverts that expectation. The most plausible explanation is contextual: in Bangladesh's insurance sector, operating under IDRA compliance requirements, data security and record integrity are perceived as mandatory baseline conditions that every system must satisfy, not as distinguishing capabilities that build readiness. When a feature is universally required, it ceases to discriminate between more- and less-ready organizations, and its statistical association with readiness collapses. This reframes system integrity from an adoption driver to a regulatory hygiene factor, a necessary precondition whose presence is assumed rather than a strategic differentiator whose enhancement motivates adoption. The distinction has direct implications for how the HOT-fit technology dimension should be theorized in tightly regulated industries, where baseline compliance requirements may neutralize the explanatory power of otherwise salient technological attributes. Ultimately, these results strongly affirm the central role of Organizational Readiness as the crucial mechanism connecting technological, human, and environmental elements to Actual Adoption within the integrated HOT-TOE framework.

5.1 Theoretical Implications

The integration of the Technology-Organization-Environment (TOE) framework and the Human-Organization-Technology fit (HOT-fit) model in this study provides a robust empirical foundation for analyzing technology adoption in the Bangladesh life insurance industry. This combined approach allows for a multidimensional assessment, capturing technological, organizational, human, and environmental factors that influence digital transformation in this sector. The empirical relevance of this integration is supported by emerging economies, where such frameworks have been shown to effectively explain technology adoption decisions in complex organizational settings [79,83,84].

A key theoretical advancement in this study is the introduction of "perceived cost" as a moderating variable within the combined TOE and HOT-fit model. Prior research consistently identifies perceived cost as a significant determinant of technology adoption, influencing managerial decisions and organizational readiness [84,85]. By empirically testing perceived cost as a moderator, this study extends the explanatory power of the combined framework, offering nuanced insights into how financial considerations shape the adoption process in resource-constrained environments like Bangladesh.

Additionally, the study introduces two novel variables which are interoperability and perceived organizational benefit within the sustainability factor of the combined model. Interoperability, or the ability of new technologies to integrate seamlessly with existing systems, is increasingly recognized as critical for successful and sustainable digital transformation yet remains

underexplored in the context of insurance and financial services [84,86]. Perceived organizational benefit, reflecting the anticipated value and performance improvements from technology adoption, further enriches the sustainability dimension and aligns with calls in the literature for more context-specific constructs [87]. By empirically validating these variables in the Bangladesh life insurance sector, this study not only advances theoretical understanding but also provides a comprehensive framework for future research on technology adaptation in similar emerging market contexts [86].

A further theoretical contribution arises from a null result. System integrity, though central to blockchain's value proposition and widely modeled as a technological driver of adoption, did not significantly predict organizational readiness in this context. This suggests a boundary condition for the HOT-fit technology dimension: in tightly regulated industries, technological attributes that are mandated as baseline compliance requirements may lose their power to differentiate more- and less-ready organizations. Rather than functioning as adoption drivers, such attributes operate as regulatory hygiene factors whose presence is assumed. This finding indicates that the explanatory weight of specific HOT-fit technology constructs is contingent on the regulatory environment, and that framework applications in compliance-heavy sectors should distinguish baseline-mandated from strategically differentiating technological features.

5.2 Empirical Implications

This represents a significant contribution to the technology adaptation literature in the insurance sector in Bangladesh by adopting a combined HOT-fit and TOE framework enriched with the sustainability factors of interoperability and business model readiness. Integrating these dimensions into one framework provides a complete lens through which our research evaluates and supports the adoption of such advanced technologies as AI and blockchain in the life insurance industry.

The inclusion of interoperability ensures that new digital solutions can easily interact with existing systems; this is very critical in minimizing disruption and maximizing efficiency. Business model readiness, on the other hand, looks at the readiness of the organization to realign its processes and value propositions. The novel use of "perceived cost" as a moderating variable by this study deepens this analysis by showing how cost perceptions could hasten or hinder technology adoption. This approach enables organizations to proactively address financial concerns, hence enhancing chances of successful implementation.

AI-driven automation in underwriting, claims processing, and customer service, together with blockchain for secure data management and transparent transactions, can help the Bangladeshi life insurance sector achieve operational excellence. Overall, these technologies contribute to reducing operational costs and processing times, enhancing accuracy, and improving customer trust. We also address the persistent barrier of stickiness to legacy.

5.3 Managerial Implications

The findings carry concrete implications for four groups: insurance executives, regulators, technology vendors, and workforce development planners. For insurance executives, the central message is that organizational readiness is the gateway to adoption, and it is buildable. The three strongest readiness predictors are Technology Readiness, Business Model Readiness, and Perceived Organizational Benefit. The practical sequence is: audit existing IT infrastructure, develop board-level business cases aligning specific AI or blockchain use cases (claims automation, fraud detection, smart contract administration) with measurable gains, then deploy. Management Support was also significant, pointing to top-down digital leadership as more effective than bottom-up experimentation. For regulators and policymakers, the exclusion of Regulatory Support from this model is itself a policy signal. Bangladesh's currently passive InsurTech regulatory stance leaves adoption drivers unrealized. Proactive InsurTech sandboxes, data governance guidelines for blockchain insurance records, and explainable-AI standards for underwriting would shift the adoption calculus for mid-tier insurers. For technology vendors, the negative effect of Stickiness to Legacy Systems signals a market preference for modular, API-first integration paths over wholesale system replacement. Perceived Cost directly predicts adoption ($\beta = 0.180$, $p < 0.001$), so phased pilots with demonstrable ROI milestones will reduce this barrier more effectively than large upfront deployments. For workforce development, Digital Literacy of Blockchain's positive effect on readiness means targeted upskilling in blockchain fundamentals and AI governance, particularly for underwriting and claims staff, will yield organizational-level readiness gains. These findings are likely to extend to other developing markets to the extent that they share these specific conditions, legacy-system inertia, limited digital literacy, and cost-sensitive organizational environments, rather than by virtue of developing-economy status alone.

6 Limitations and Future Research

A primary limitation is the cross-sectional design. All variables were measured at a single point in time, meaning the structural paths reported here reflect statistical associations, not causal sequences. The language throughout this manuscript uses “predicts” and “is associated with” deliberately; establishing causal direction requires longitudinal or experimental designs that this study does not provide. Panel data tracking the same organizations across multiple periods would allow examination of how readiness evolves with technology maturity, whether adoption follows from readiness improvements over time, and what post-adoption outcomes, including user satisfaction, utilization rates, and performance gains, as they evolve longitudinally. Extending the HOT-fit model temporally to include these post-adoption dimensions remains an important direction for future research [78]

Second, the empirical scope was limited exclusively to the life insurance industry in Bangladesh, which inherently constrains the generalizability of the findings to other contexts. The variable “regulatory support,” which could typically be classified under the sustainability factor, was excluded from the theoretical framework. This exclusion was necessitated by the current lack of technological advancement within the Bangladeshi regulatory environment. If the regulatory body were technologically proactive, it would provide a sufficient impetus to convince insurance

companies to implement AI and blockchain technology. The results may not directly apply to developed economies or other financial sectors where technological infrastructure and organizational culture differ. More importantly, transferability should be treated as conditional on contextual similarity rather than on shared developing-economy status. The findings are most likely to hold in settings that resemble Bangladesh on three dimensions: a regulatory environment where technology and data-security standards operate as compliance baselines rather than incentivized differentiators; an insurance sector at an early, pre-scaled stage of InsurTech maturity where AI and blockchain remain largely at pilot or evaluation stage; and a workforce whose digital literacy in emerging technologies is still developing. Economies differing on these dimensions, such as those with proactive InsurTech regulatory sandboxes or mature digital-insurance markets, may exhibit different readiness dynamics. Future research should therefore test the model in contexts selected for their similarity to or contrast with these features, rather than assuming transfer across developing economies as a category.

Although multiple diagnostics (Harman's single-factor test, full collinearity VIF, and a marker-based adjustment) indicated that common method variance is unlikely to have materially affected the results, all data were collected from single respondents at a single point in time using one instrument. Post-hoc statistical remedies cannot fully rule out common method bias, and the marker employed was ad-hoc rather than designed a priori. Future studies would benefit from temporal or source separation of predictor and criterion measures, or the inclusion of a purpose-designed marker variable.

A further consideration concerns the level of analysis. Organizational constructs were measured through individual employee perceptions and modeled at the individual level rather than aggregated to the firm level. While perception-based measurement is standard in survey-based adoption research, it means the findings describe how individual employees perceive organizational readiness and its drivers, not objective organizational states verified through firm-level or archival data. Claims about organizational-level adoption should therefore be read as reflecting aggregated individual perceptions. Future research could employ multi-level designs, collecting multiple respondents per firm with inter-rater agreement statistics to justify firm-level aggregation and enable genuinely organizational-level inference.

7 Conclusion

This research's primary goal was to determine the factors influencing AI and Blockchain adoption in life insurance sector by creating an integrated structural model combining the Human-Organization-Technology (HOT-fit) and Technology-Organization-Environment (TOE) frameworks. The study specifically aimed to investigate how human, technological, organizational and Sustainability factors collectively shape organizational readiness, and subsequently, how readiness drives actual adoption. A key secondary objective was to examine the moderating role of perceived cost, recognizing the financial constraints typical in developing economies.

The study conclusively demonstrated that perceived organizational readiness is the most powerful predictor of employees' reported adoption. This readiness is associated with several key antecedents, including technology readiness, business model readiness, management support, digital literacy of blockchain, interoperability, and perceived organizational benefit. Conversely, stickiness to legacy systems showed a significant negative association with readiness, suggesting technological inertia may impede adoption. Furthermore, the moderating effect of perceived cost was substantial, confirming that even when an organization is fully prepared, high perceived expenses can temper the likelihood of adopting emerging technologies.

Theoretically, this research advances the literature by empirically validating the combined HOT-TOE framework, showing that readiness is a complex, multidimensional construct shaped by both internal capabilities and external conditions. This work also extends these models by proving that cost and sustainability factors significantly refine their explanatory power. Practically, the findings offer clear guidance for insurers and policymakers to enhance readiness through strategic alignment, dedicated managerial support, workforce upskilling, and effective cost management. The study's most theoretically notable result was a null one: system integrity did not significantly predict organizational readiness, despite being blockchain's defining feature. In a compliance-driven market, security appears to function as an assumed baseline rather than a readiness-building differentiator, a finding that cautions against treating technological assurance as a universal adoption driver across regulatory contexts.

Ultimately, the study contributes to both theory and practice by establishing organizational readiness as the essential bridge between technological intention and successful digital implementation. For practitioners, the findings point toward specific, actionable levers: strengthen technology infrastructure, build management commitment, invest in digital literacy, and actively manage the cost perception associated with new technology programs. Future work should extend these findings longitudinally, replicate the model in other emerging market insurance contexts, and disentangle the separate adoption dynamics of AI and blockchain as distinct technologies.

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